MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TWO SHIELDED TWISTED PAIRS NO. 16 GAUGE, FLUOROPOLYMER INSULATION, FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 2SPR-16

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, tin coated, AWG size 16, class B stranding.
Second - Insulation of ethylene tetrafluoroethylene (TEFZEL). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.
Third - Two insulated conductors, one black and one white, twisted together with a lay of 1-1/4 ± 1/8 inch.
Fourth - Braided shield of AWG no. 36 tin coated copper. Braid angle of 35 degrees minimum; coverage of 85 percent, minimum.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.
Sixth - Two shielded pairs cabled with a lay not greater than 24 times the pitch diameter of the layer.
Seventh - Rubber fillers in order to form a firm, round assembly. Fillers shall be capable of removal from the assembly without damaging the shield insulation.
Eighth - Polyester binder tape applied helically with overlap.
Ninth - Cable jacket of black polyether polyurethane. The cable jacket material shall be B.F. Goodrich Co. ESTANE 58300 or 58863; or Mobay Chemical Co. TEXIN 985A. Nominal wall thickness 0.110 inch. Overall cable diameter to be 0.630 to 0.650 inch.

FSC 6145
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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - at 20°C, ohms/1000 feet, maximum.	4.24
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield	2000
Shield to water	500
Insulation dielectric (on 100 percent of insulated conductor) - test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum.	
Insulation resistance - megohms/1000 feet, minimum	5000
Group A:	
Visual and dimensional	No failure
Group B:	
Cold bending, cable - at minus 40 ± 2°C over a 4-inch diameter mandrel	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	5000
Elongation - percent, minimum	100
Cable jacket	
Tensile strength - lbf/in^2, minimum	5000
Elongation - percent, minimum	500
Group C:	
Physicals (aged)	
Insulation	
Air oven (168 hours at 180°C)	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	90
Cable jacket	
Air oven (168 hours at 80°C)	
This test shall be in accordance with MIL-C-915	No failure
Hot oil immersion	
Tensile strength, percent of unaged, minimum	90
Elongation, percent of unaged, minimum	80
Shield-conformance to material coverage and construction	No failure

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QUALIFICATION INSPECTION:
Qualification inspection shall include basic electrical, plus all of groups A, B and C.

UNIT ORDERING LENGTH: 500 feet (minimum).

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0886-04)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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